Security incident report

Section 1: Identify the network protocol involved in the incident
The network protocols involved in this incident are **DNS (Domain Name System)** and **HTTP (Hypertext Transfer Protocol)**. • **DNS** was used by the user's machine to resolve the domain name yummyrecipesforme.com to its corresponding IP address (203.0.113.22). • **HTTP** was used by the browser to establish a connection to the web server on port 80 and request the webpage (HTTP: GET / HTTP/1.1). The logs also show a second DNS request for greatrecipesforme.com and a subsequent HTTP connection to the spoofed IP address (192.0.2.172), which is where the malicious file redirected the user.

Section 2: Document the incident
Incident Summary: An ex-employee gained unauthorized access to the website's administrative panel by performing a brute-force attack using default passwords. After gaining access, the attacker injected malicious JavaScript code into the website's source code and changed the administrator password, locking out the legitimate owner. **How it was discovered:** The incident was discovered after multiple customers contacted the help desk complaining that they were prompted to download a file to access free recipes. Customers reported that after running the file, their browser redirected to a different website (greatrecipesforme.com), and their computers began running slowly.

Investigation and Evidence:
The investigation was conducted by cybersecurity analysts using a sandbox environment. By running tcpdump and visiting the URL, analysts observed the HTTP request for yummyrecipesforme.com and the subsequent download of an executable file. The tcpdump logs show the browser making a DNS request for greatrecipesforme.com and connecting to its spoofed IP address (192.0.2.172).
A senior analyst confirmed the compromise by reviewing the website's source code and finding the injected JavaScript.

Root Cause:
The attack was successful because the administrative account was still using a default password, and there were no security controls (such as account lockout policies or MFA) to prevent brute-force attempts.

Section 3: Recommend one remediation for brute force attacks
Recommendation: Implement **Multi-Factor Authentication (MFA)** for all administrative accounts. **Why it is effective:** MFA requires a user to provide two or more verification factors to gain access (e.g., a password *and* a code sent to their phone). Even if an attacker successfully guesses the password through a brute-force attack, they will still be unable to access the admin panel without the second factor. This provides a critical layer of defense for high-privilege accounts. *(Note: Enforcing strong password policies and an Account Lockout policy after a certain number of failed attempts are also highly effective measures that should be implemented alongside MFA).*